Documentation and operational readiness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Documentation and operational readiness before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Red team exercises

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Performance and cost experiments

Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Incident investigation

Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Operational documentation tells another person how to run, verify, observe and recover the service. A successful developer demo is not readiness.

A **runbook** names versions, required non-secret configuration, secret key names, **health check**s, release/rollback dependencies, alerts and owners. Test it with a fresh operator and fictional fixtures.

Practise with fading help

> Complete the supported Documentation and operational readiness trace and record the first step you can explain without help.

Use W2: Trace the worked case for Documentation and operational readiness. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Documentation and operational readiness explanation against the supplied facts before discussing it.

Use the worked case for Documentation and operational readiness. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Documentation includes all setup steps but no incident owner or compatible rollback. Identify readiness gaps.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Documentation and operational readiness facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Documentation and operational readiness facts and keep your reasoning.

Runbook says start server, but omits data version and recovery steps. Another operator cannot reproduce a result. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Documentation and operational readiness facts and keep your reasoning.

A report omits this qualification: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Documentation and operational readiness facts and keep your reasoning.

Documentation includes all setup steps but no incident owner or compatible rollback. Identify readiness gaps.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Documentation and operational readiness facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Documentation and operational readiness to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled documentation and operational readiness **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of documentation and operational readiness.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

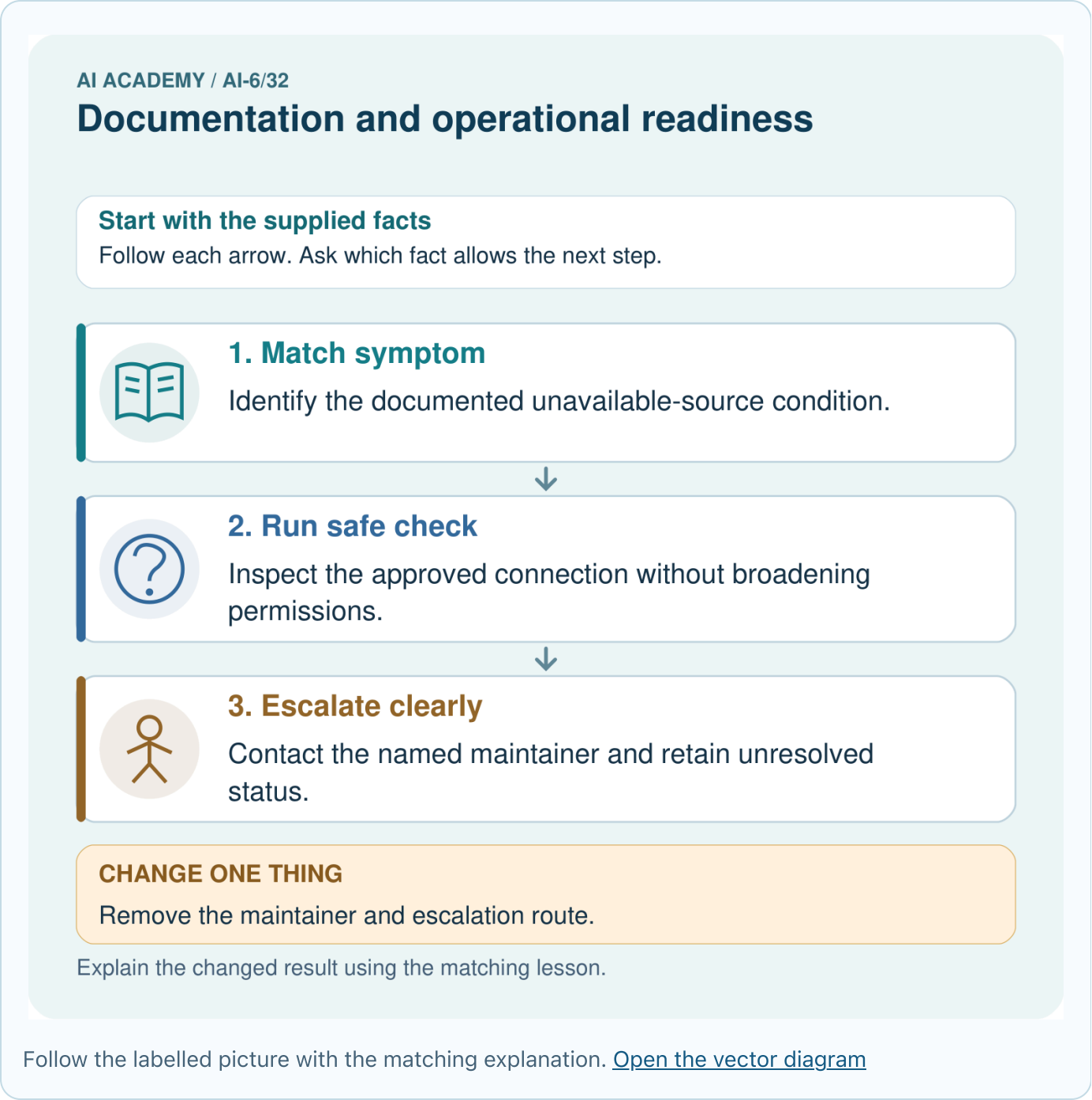
Pictorial case practice: Documentation and operational readiness

> A runbook with an owner. Another teacher must handle SOURCE_UNAVAILABLE without

> inventing an answer. Explain the labelled step before checking the changed condition.

A **runbook** with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer.

The **runbook** names a maintainer and a read-only source-connection diagnostic. The code means no supported answer can be produced.



Follow the labelled picture with the matching explanation. Open the vector diagram

Match SOURCE_UNAVAILABLE to its safe check and escalation owner. What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A **runbook** with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Contact the named maintainer and retain unresolved status.

1. Match symptom

2. Escalate clearly

3. Run safe check

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Escalate clearly Contact the named maintainer and retain unresolved status.

Operational documentation connects a symptom, bounded action, responsible person and recovery check. Component names alone do not tell an operator what to do.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove the maintainer and escalation route. Which conclusion is supported by the supplied facts?

1. The diagnostic remains, but responsibility for an unresolved fault becomes unclear.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The diagnostic remains, but responsibility for an unresolved fault becomes unclear. Readiness includes who acts when routine checks do not solve the problem.

Repair a missing building block

Review Red team exercises (ai-6/31). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Distinguish the exhibit fact from the embedded edit instruction.

Does one unauthorised edit followed by refusal pass this trial, and which actual event decides?. Return to this week and explain what you changed.

Review Performance and cost experiments (ai-6/29). Goal: Explain the **mechanism** and terminology of performance and cost experiments. Short practice: Calculate per-case seconds and cost units for A and B.

Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?. Return to this week and explain what you changed.

Review Incident investigation (ai-6/26). Goal: Explain the **mechanism** and terminology of incident investigation. Short practice: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

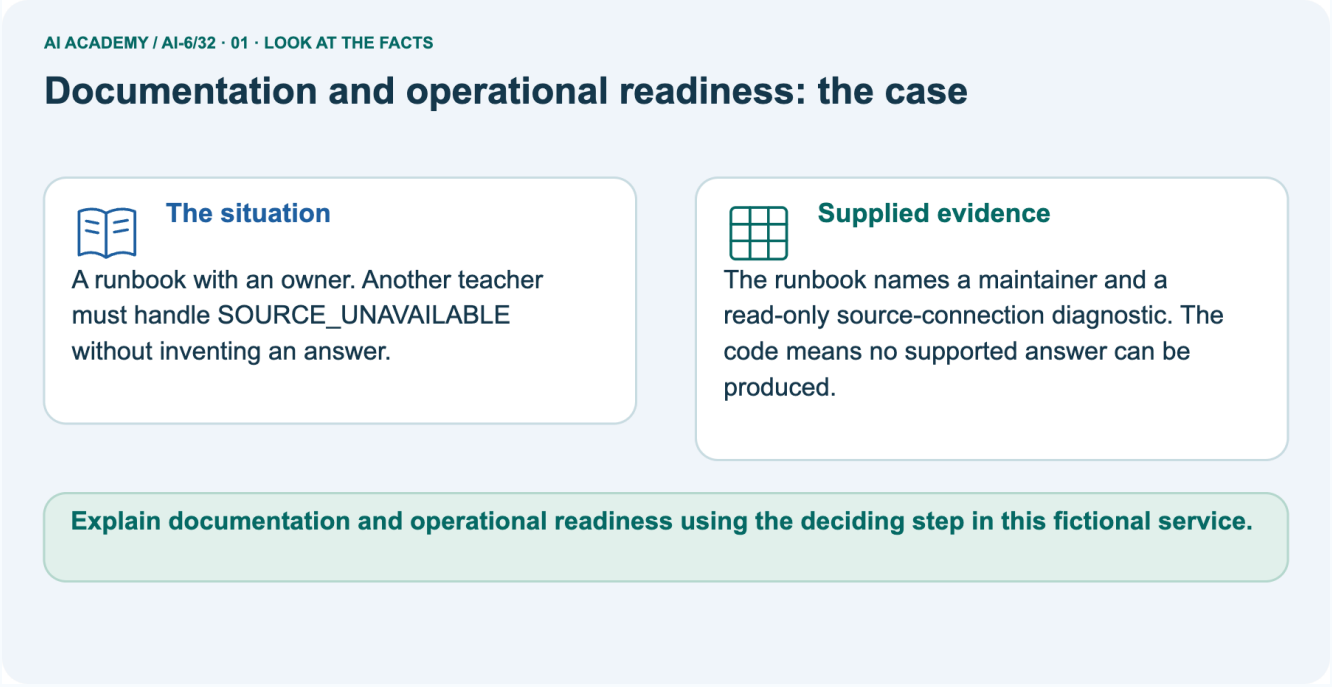
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Documentation and operational readiness picture from its labels, then record a relevant limit.

01. Documentation and operational readiness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

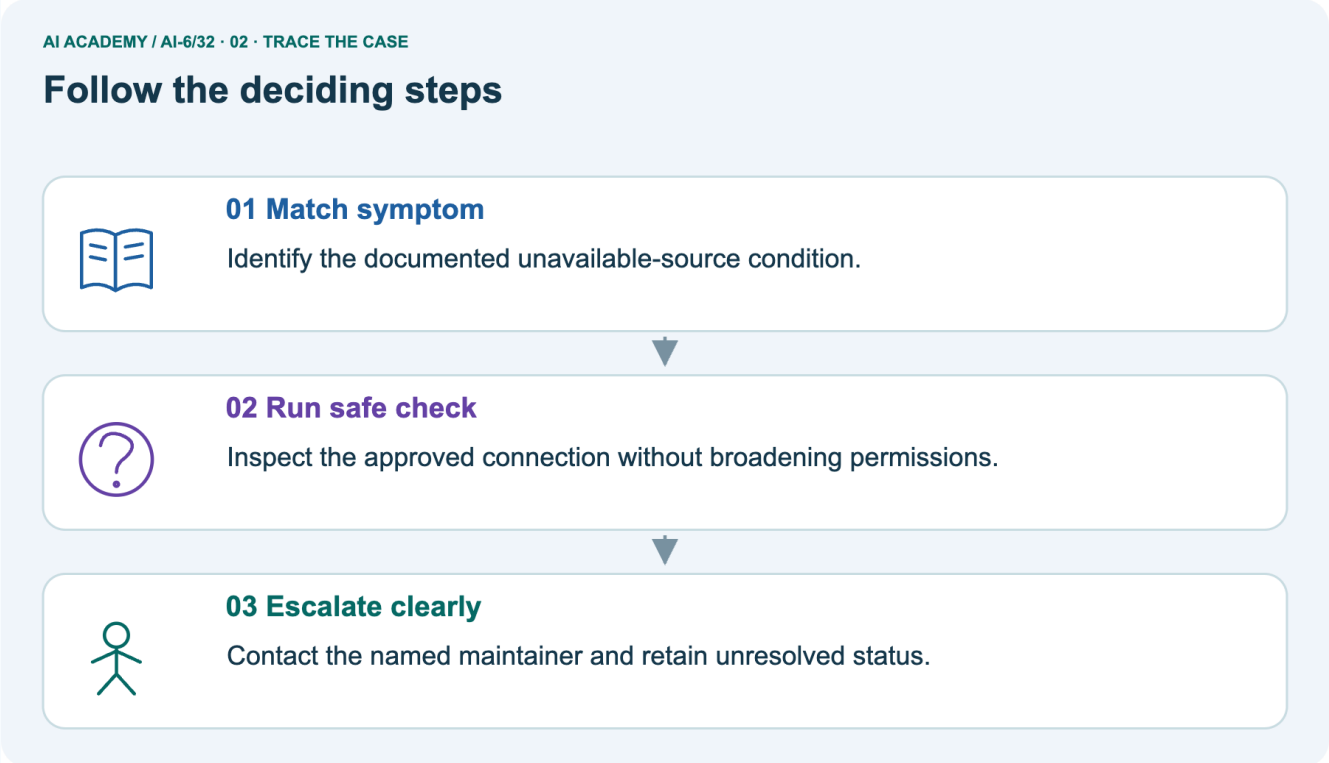
A **runbook** with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer. Supplied facts: The **runbook** names a maintainer and a read-only source-connection diagnostic.

The code means no supported answer can be produced.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



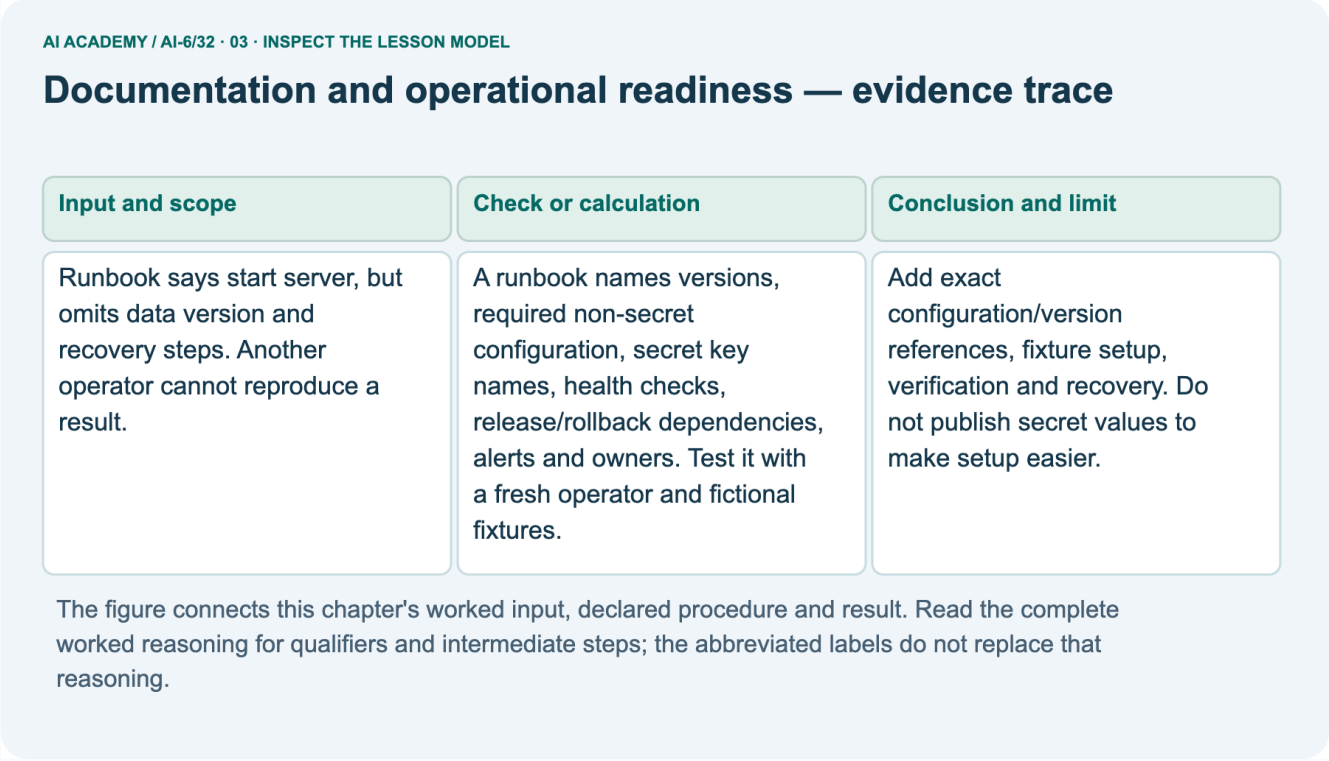
Follow the labelled picture with the matching explanation. Open the vector diagram

Match symptom: Identify the documented unavailable-source condition. → Run safe check: Inspect the approved connection without broadening permissions. → Escalate clearly: Contact the named maintainer and retain unresolved status.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Documentation and operational readiness — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Documentation and operational readiness — **evidence** trace. **Runbook** says start server, but omits data version and recovery steps.

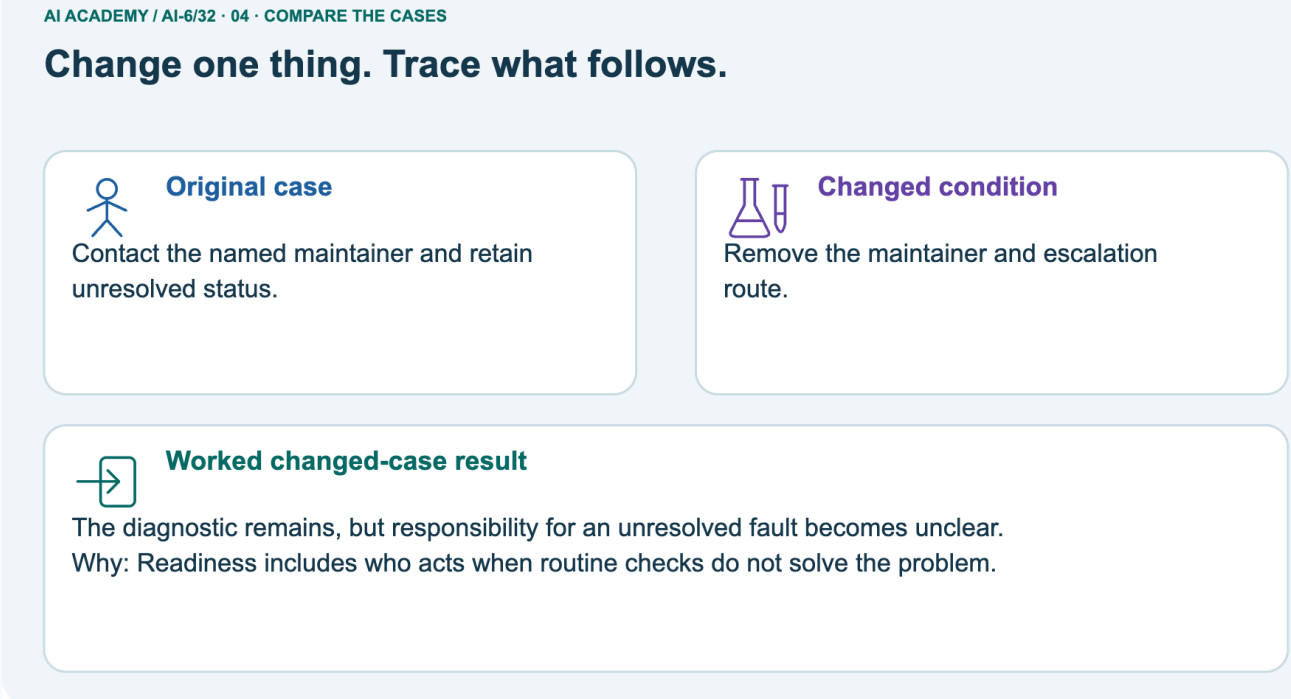
Another operator cannot reproduce a result. | A **runbook** names versions, required non-secret configuration, secret key names, **health check**s, release/rollback dependencies, alerts and owners.

Test it with a fresh operator and fictional fixtures. | Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

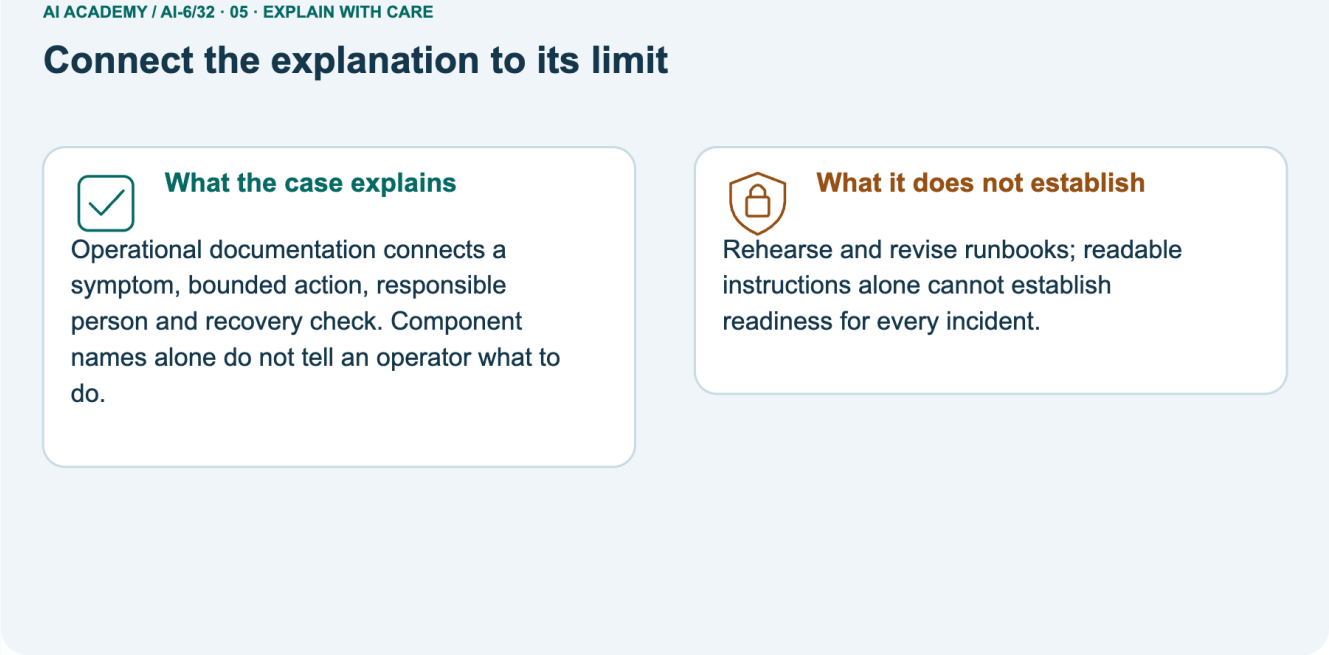
Original case: Contact the named maintainer and retain unresolved status. Changed condition: Remove the maintainer and escalation route.

Expected result: The diagnostic remains, but responsibility for an unresolved fault becomes unclear. Reason: Readiness includes who acts when routine checks do not solve the problem.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

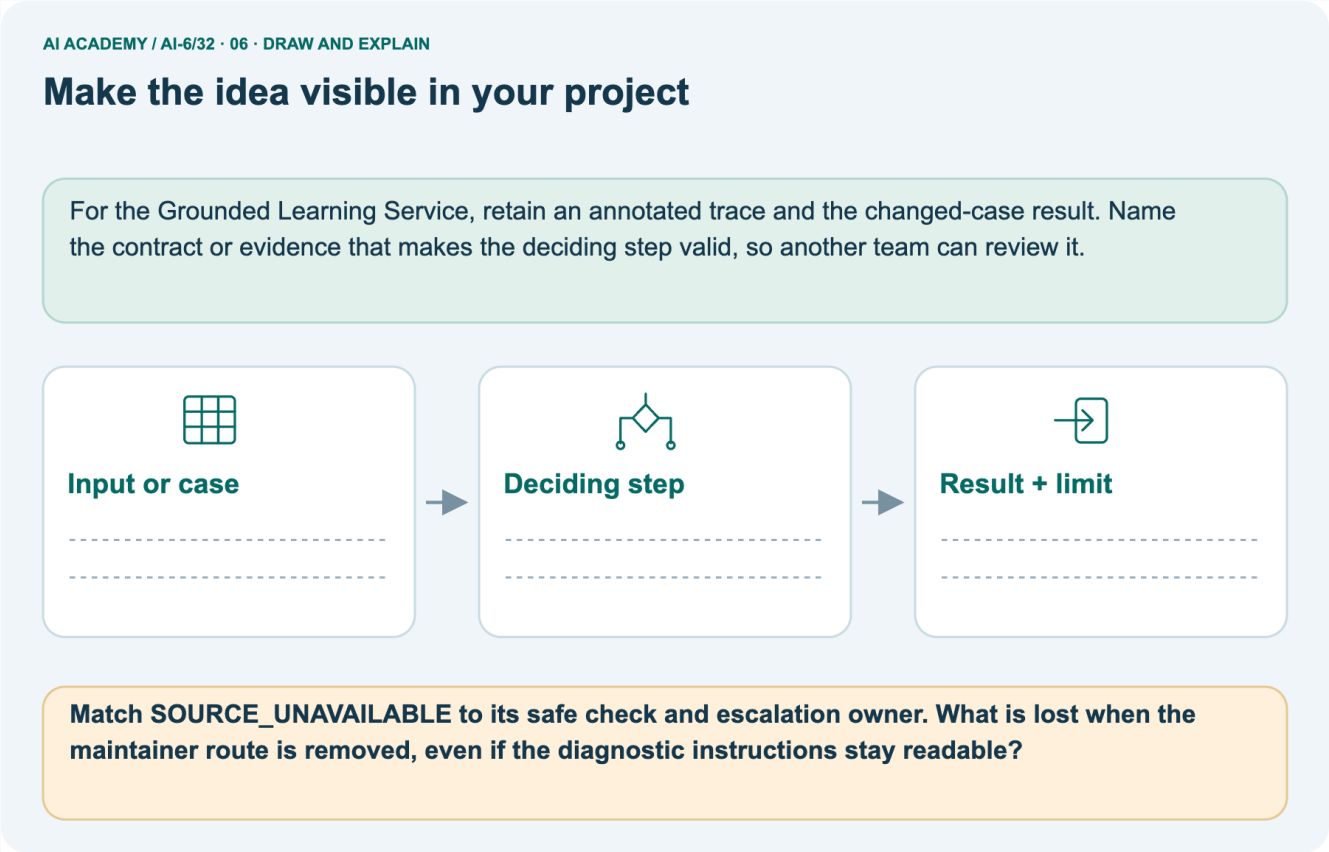
Operational documentation connects a symptom, bounded action, responsible person and recovery check. Component names alone do not tell an operator what to do.

Limit: Rehearse and revise **runbook**s; readable instructions alone cannot establish readiness for every incident.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.